

PREVENTION OF HEALTHCARE-ASSOCIATED LEGIONELLA DISEASE AND SCALD INJURY FROM POTABLE WATER DISTRIBUTION SYSTEMS

SOP 138-21-05

VA Palo Alto Health Care System
Palo Alto, CA 94304

Service Line:
Engineering Service

Signatory Authority:
Chief, Engineering Service

Effective Date:
December 15, 2021

Responsible Owner:
Chief, Engineering Service

Recertification Date:
December 31, 2026

1. PURPOSE AND AUTHORITY

- a. This standard operating procedure (SOP) establishes guidelines for prevention of Healthcare-Associated Legionella Disease (HCA LD) and Scald Injury from Potable Water Distribution Systems in VA Palo Alto Health Care System (VAPAHCS) buildings and in which patients, residents, or visitors stay overnight.
- b. VAPAHCS will take steps in accordance with VHA Directive 1061, Prevention of Healthcare-Associated Legionella Disease and Scald Injury from Potable Water Distribution Systems to prevent and control HCA LD. Buildings covered by this SOP include: Palo Alto Campus: 100, 7, 520, MB2, 8, 9, 500, Menlo Park Campus: 347, 348, 349, 350, 351, 352, 360, 331, 365, and the Livermore Campus: 90.

2. PROCEDURE

- a. Engineering Service will develop a written Healthcare-Associated Legionella Disease (HCA LD) Management Plan with specific procedures for each building subject to VA Directive 1061.
- b. Engineering control strategies and limits for ongoing prevention of Legionella growth will be developed and implemented as follows:
 - (1) Water Quality and Pressure Monitoring: Potable water entering each building subject to this Memorandum shall be continuously monitored for incoming water pressure and the following characteristics: temperature, pH, dissolved solids, and oxidant residual.
 - (2) Water Temperature. VHA Directive 1061 requirements for water temperature limits for Legionella control in the building's potable hot and cold-water distribution systems are summarized as follows:
 - (a) Hot Water Distribution Systems: In buildings with domestic hot water storage tanks, water temperature for storage tanks must be maintained at

a minimum of 140 degrees Fahrenheit (°F) (60 degrees Celsius (°C)) to prevent Legionella growth. The minimum discharge temperature for instantaneous and semi-instantaneous heat exchangers must be 130°F (54.4°C). Water in the potable hot water distribution system piping must be no lower than 124°F (51.1°C). 124°F (51.1°C) and the temperature will be maintained prior to any temperature-reducing mixing valve or anti-scald device at the water outlet.

(b) Cold Water Distribution Systems: Legionella can grow in the building's cold water distribution system as water temperatures increase above 67°F (19.4°C). Cold water temperature throughout the system should be maintained at or below 67°F (19.4°C) to the greatest extent practicable to inhibit growth.

(3) Water Temperature Monitoring. Water temperatures in the hot and cold potable water distribution systems will be monitored continuously to determine if temperatures are within the established control limits. Temperature monitoring must be conducted, at a minimum, at the following locations: incoming water supply to the building, water storage tanks, hot water discharge at the hot water source equipment, hot water return proximal to the hot water source equipment, water at the return of circulation loops, and water supplied to representative outlets (e.g., loop or branch, hydraulic remoteness, flow).

(4) Water Temperature Control at the Outlet. Efforts will be taken to minimize risk of scald injury to patients, residents, staff and visitors in buildings covered by this SOP. Mixing valves and anti-scald devices will be placed on water outlets (plumbing fixtures) where people access water from the potable hot water distribution system in order to prevent scald injury. The water temperature delivered from the outlet must not exceed 110°F (43.3°C).

(a) Mixing valves shall be set at 107°F (41.6°C) to account for temperature range tolerances of the valves. Setting the valves at 107°F (41.6°C) will ensure the outlet temperature will not exceed 110°F (43.3°C).

(5) Biocide. Oxidant residual levels in the building incoming water supply and at representative outlets must be assessed. Assessments will determine if any disinfectant water treatment from the municipality or other potable water source is present when water reaches the building and after distribution within the building. Knowing oxidant values will aid in determining if oxidant residual levels are at a sufficient level to suppress Legionella growth (if present) and will contribute to the information available if deciding whether or not to install additional treatment system(s).

(a) Monitoring of oxidant residual levels in the incoming water supply is to be continual. Additionally, monitoring oxidant residual in water supplied to

representative outlets (e.g., loop or branch, hydraulic remoteness, flow) is to be continual.

(b) Minimum concentrations of oxidant residual necessary for inhibition of *Legionella* growth may vary from building to building. In general, the following minimum detected oxidant residual levels at hot and cold-water outlets are suggested as guidance: 0.5 milligrams (mg) per liter (L) for chlorine (as free chlorine), 0.5 mg/L for monochloramine, and 0.3 mg/L for chlorine dioxide.

(c) The water treatment electronic ionization systems for Buildings 100, 7, 90, 500, and 520 will be set to maintain a range of 0.2-0.4 parts per million (PPM) Copper and 10-60 PPM Silver. Records of temperatures and copper/silver ranges will be maintained by Engineering Service (138C5). Temperatures and copper/silver ranges that fall above or below the mandated requirements will be reported to the Water Safety committee immediately through the Chief of Engineering Service.

(8) Flushing. Regular flushing of hot and cold water at outlets (e.g., sink taps, showers) will be done for outlets not in routine use or which experience low water flow. Irregular use or low flow fixtures must be flushed at least twice per week to prevent water stagnation for extended periods of time. Flushing will be done by allowing the fixture to run wide open for five minutes on the hot water outlets.

(9) Corrective Actions. If routine monitoring determines water temperatures or biocide residual levels from an installed system are not within the established limits, then the following actions, at a minimum, must occur:

(a) Assess the reason why the water systems controls were not within established limits.

(b) Corrective actions must be undertaken promptly, based on the assessment, to satisfy implementation of the water system control measures within established limits.

(c) Re-assess controls measures after corrective actions are implemented to determine if the water system management point is within established parameters. If not within established parameters, reassess the corrective actions, and implement revised corrective actions.

(10) Documentation. Water temperature and biocide residual testing, as well as corrective actions will be documented to provide verification of implementation and monitoring. Records of this water testing will be maintained by the Water Safety Committee.

- c. Environmental Validation: For this SOP, environmental validation is testing the building's potable water distribution system(s) to determine if the engineering

controls are successfully inhibiting growth of Legionella. This section provides requirements and recommendations for water testing and describes mitigation actions.

(1) Testing of VAPAHCS buildings hot and cold-water distribution systems for *L. pneumophila* will be performed quarterly (once per Federal Fiscal Year quarter). Water samples from at least 10 randomly selected outlets on the hot water distribution system and at least 10 randomly selected outlets on the cold-water distribution system will be tested from each building for each quarterly testing cycle. At the time each water sample is taken, test and document the following for each sample: water temperature, level of residual biocide, and pH.

(2) Once collected, the same day samples are to be shipped to the reference lab. The Water Safety Committee is to select the testing laboratory using criteria outlined in VA directive 1061.

(3) If environmental testing detects *L. pneumophila* then the actions outlined below will be instituted. Any amount of *L. pneumophila* detected in a sample is considered a positive result. The following actions will be taken by the Water Safety Committee:

(a) Assess implementation of the engineering controls (e.g., water temperature, biocide levels, etc.) to determine if corrective adjustments need to be taken.

(b) Implement remedial action in the HCA LD Prevention Plan using the criteria below. This remedial action approach uses a "graded response" for addressing *L. pneumophila*-positive samples detected through routine water testing. That is, while each positive sample will require further assessment, the extent of remediation is situation dependent.

(c) If one outlet is positive for *L. pneumophila*, then a determination must be made as to which other outlets in the area (on same water distribution loop) must be tested for *L. pneumophila* and if any mitigation of the immediate water distribution loop or area is necessary pending results of any additional cultures. The plumbing fixture testing positive for *L. pneumophila* must be promptly remediated.

(d) If more than one outlet is positive for *L. pneumophila*, assess the results to determine the subsequent actions as follows:

(e) If water outlets testing positive are in the same area of the building or on the same water distribution loop, then promptly conduct remediation of the area or loop.

- (f) If water outlets testing positive are in different areas of the building or on different water distribution loops, then promptly conduct remediation in a graded response based on location of the positive outlets. The Water Safety Committee will meet to review the location of the positive water samples in relation to the configurations of the building water distribution system(s) to determine the extent of remediation (i.e., areas/loops versus entire building).
- (4) After environmental remediation is completed, prompt re-testing of the areas testing positive for *L. pneumophila* will be done to determine if remediation procedures were successful at reducing *L. pneumophila* to undetectable levels. If the remediation procedures were successful, the quarterly water environmental validation cycle is complete. If remediation procedures did not reduce *L. pneumophila* to undetectable levels, the Water Safety Committee will assess post-remediation results for location of positive samples, and determine subsequent actions based on the guidance in this section.
- (5) Documentation is required for environmental testing (date, outlets, and results), any assessments of positive results, any remedial action taken, and efficacy of remedial actions. A report of environmental testing is to be submitted to the Water Safety Committee, Environment of Care Committee (EOCC), and Infection Control Committee.
- (6) If environmental water testing detects *Legionella* but the *Legionella* is not *L. pneumophila*, the following actions are required since growth of non-*pneumophila* *Legionella* species indicates conditions are conducive for growth of *L. pneumophila*.
- (a) Assess engineering controls (e.g., water temperature, biocide levels) and flushing protocols and adjust, as necessary, for *Legionella* control.
- (b) Further testing for *L. pneumophila* of the water distribution systems is required to determine if *L. pneumophila* is present. Testing should be focused on areas where positive water samples were identified.
- (c) If *L. pneumophila* is found, follow procedures outlined above in this SOP.
- (d) If *L. pneumophila* is not found, the quarterly environmental validation cycle is complete.
- d. Emergency remediation of a building's potable water distribution system(s) is triggered by certain occurrences: identification of a definite HCA LD case, identification of a possible HCA LD case and *Legionella*-positive water results, or

identification of Legionella-positive water results during routine environmental testing.

- e. Emergency remediation will be done in a step wise approach and includes any or all the following procedures:

(1) Sanitizing and flushing of the fixture: If a positive result is returned on routine testing, the first remediation step is plumbing fixture sanitizing. This is accomplished by replacement of corroded and calcified areas. When replacement parts are installed, components taken apart are sanitized with Cavi1Wipes. The fixture is then flushed for five minutes. In addition, plumbing systems in the immediate area will be examined for presence of dead legs and if found will be removed. Consideration should be given to installing a replaceable, in-line, FDA-cleared, 0.22-microns pore size filter and removing aerator and laminal flow devices if present.

(2) Thermal Eradication: This procedure involves temporary resetting of the temperature in the hot water distribution system(s) to 160 °F - 170°F (71°C - 77°C) while continuously flushing each water system outlet for at least 30 minutes. Consideration needs to be given as to the feasibility of implementing thermal eradication depending on the mixing valve designs in place.

(3) Shock chlorination: This method involves increasing chlorine levels of hot and cold water distribution systems to at least 2 mg/L and maintaining that level throughout the water systems for at least 2 hours (but not exceeding 24 hours) and flushing outlets. Chlorination of the hot water tank(s) or water heater(s) to a concentration of 20 to 50 mg/L may be required to achieve this level of free chlorine residual. After shock chlorination procedure is complete, the system must be thoroughly flushed before re-use. If post-shock chlorination water testing indicates Legionella bacteria are still present in the water distribution system(s), it may be necessary to repeat shock chlorination with consideration for use of a higher concentration of chlorine (e.g., at least 10 mg/ml free chlorine residual throughout the water system and at outlets for 24 hours or 200 mg/L for three hours (refer to disinfection definition).

(4) Thermal eradication and shock chlorination are temporary measures. After emergency mitigation, environmental testing will be performed to determine effectiveness of the mitigation action.

(a) Prior to implementation of emergency mitigation, stakeholders must be informed the process will take place to facilitate safe implementation of the emergency procedures. After the mitigation process is complete, communication must occur to inform stakeholders that the water is acceptable for general use. The Water Safety Committee must document any emergency mitigation processes that take place.

(b) It is important to consider implications of Legionella mitigation strategies on special use water systems within the building. For example, chemical disinfectants may result in the introduction of products into, or the formation of disinfection byproducts in, the building water supply at concentrations that may be toxic to patients on hemodialysis. Employees responsible for the oversight of special use water systems are to be consulted during the development and implementation of water treatment strategies for Legionella and promptly notified of any changes in treatment procedure.

- f. Engineering Service will maintain HVAC system cooling towers and other non-potable water systems using the VAPAHCS HCS LD Management Plan. Maintenance, cleaning, and monitoring of cooling tower will follow procedures in the management plan.

4. RESPONSIBILITIES

- a. The Chief of Engineering Service shall VAPAHCS compliance with the latest edition of VHA Directive 1061.

5. DEFINITIONS

- a. Continual or Continuous: Continual or continuous refers to the almost uninterrupted monitoring and control of: water quality, incoming water pressure, biocide levels, and water temperatures. This can be achieved using automated measurement/control devices, typically connected to the Building Automation System (BAS), at various locations such as water source equipment and throughout the water distribution system.
- b. Dead leg: A length of pipe in the water distribution system that is open at one end to the system and the other end terminates at a cap, blind flange or closed valve.
- c. Mixing Valve. Mixing valve is a generic reference to a class of water tempering devices. Mixing valves used for tempering hot water in potable water systems must meet the requirements of the International Plumbing Code (IPC 2009), American Society of Sanitary Engineers (ASSE 1016/1069/1070), and VA Master Construction Specifications.
- d. Outlet. Outlet is a point in the potable water distribution system where an individual (also known as the "end user") accesses the water. Examples include faucets, showers, ice machines, and drinking fountains.
- c. Shock Chlorination. Shock chlorination is the application of hypochlorite, usually in the form of a solution, to the water distribution system at higher than normal levels for remediation purposes.

6. REFERENCES

- a. VHA Directive 1061, Latest Edition

7. REVIEW

This SOP must be reviewed at minimum at recertification and including when there are changes to service-level privacy assessments.

8. RECERTIFICATION

This SOP is scheduled for recertification on or before the last working day of December 2026. In the event of contradiction with national policy, the national policy supersedes and controls.

9. SIGNATORY AUTHORITY

JAMES P. Digitally signed by JAMES P.
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James P. Crowe
Chief of Engineering Service
Date Approved: 15 Dec 2021

NOTE: *The signature remains valid until rescinded by an appropriate administrative action.*

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